Functional Requirements Document (FRD)

Hi Ecommerce – Product Return Management System

1. Document Information

Item	Details
Document Name	Functional Requirements Document
Project Name	Hi Ecommerce
Project Type	IT Business Analyst Portfolio Project
Methodology	Agile Scrum
Document Version	1.0
Document Status	Draft
Prepared By	Business Analyst
Company	Hi Ecommerce (Fictional Business)
Related Document	Business Requirements Document (BRD)

2. Introduction

2.1 Purpose

The purpose of this Functional Requirements Document is to define the functional behaviour of the proposed Hi Ecommerce Product Return Management System.

It describes how customers will interact with the system to check product return eligibility, submit return requests, upload supporting images, receive confirmation, and view return request details.

This document will guide the development team, QA testers, and business stakeholders during system design, development, and testing.

2.2 Project Overview

Hi Ecommerce is a fictional e-commerce business planning to improve its product return management process.

The proposed system will provide a structured process for customers to submit and track product return requests.

The system will also capture structured return information to support future business analysis and reporting.

2.3 Project Objectives

- Simplify the product return process.
- Improve customer experience.
- Reduce manual effort in return request handling.
- Provide clear return eligibility information.
- Maintain accurate return request records.
- Allow customers to track submitted requests.
- Capture return reasons for business reporting.

3. Scope

3.1 In Scope

- Customer authentication.
- Customer order history.
- Product return eligibility validation.
- Return reason selection.
- Return request submission.
- Supporting image upload.
- Unique return request ID generation.
- Return request confirmation.
- Return request history and details.
- Return request status display.

3.2 Out of Scope

- Real payment gateway integration.
- Automated refund processing.
- Courier and logistics integration.
- AI-based return prediction.
- Live integration with an existing e-commerce platform.
- Actual production deployment.

4. Users and Actors

Actor	Description
Customer	Logs in, checks eligibility, submits and views return requests.
Return Management Team	Reviews return requests through a future administrative workflow.
Customer Support	Assists customers with return-related queries.

Team	
System	Performs validations, generates request IDs, stores information and sends notifications.

5. Functional Requirements

FR-01: Customer Authentication

Description: The system must allow registered customers to access their accounts securely.

Requirements:

1. The system must provide a login screen.
2. The customer must enter valid login credentials.
3. The system must validate the submitted credentials.
4. The system must allow access only after successful authentication.
5. The system must display an appropriate error message for invalid credentials.
6. The system must provide a logout option.

Expected Result: Only authenticated customers can access their account and return request functionality.

Priority: High

FR-02: View Order History

Description: The system must allow customers to view their previously purchased products.

Requirements:

1. The customer must be able to access their order history after login.
2. The system must display available order references.
3. Each order must display relevant information, including order date and order status.
4. The customer must be able to view products associated with an order.
5. The system must display the product name and relevant purchase details.
6. The system must display a message when no order history is available.

Expected Result: Customers can identify the products they want to return.

Priority: High

FR-03: Check Product Return Eligibility

Description: The system must validate whether a purchased product is eligible for return.

Requirements:

1. The customer must be able to select a purchased product.
2. The system must retrieve the relevant order and product details.
3. The system must check the product against the configured return policy.
4. The eligibility validation must consider applicable rules, such as purchase date, product category, order status and product-specific restrictions.
5. The system must display an eligible or not eligible result.
6. If a product is not eligible, the system must display an explanation where available.
7. The system must prevent customers from submitting return requests for ineligible products.

Expected Result: Customers can identify whether a product can be returned before submitting a request.

Priority: High

FR-04: Select Return Reason

Description: The system must allow customers to select a reason for returning a product.

Requirements:

1. The system must display a list of available return reasons.
2. The customer must select one reason before submitting the request.
3. Example reasons include:
    - Product damaged.
    - Wrong product delivered.
    - Product defective.
    - Product not as described.
    - Size or fit issue.
    - Product quality issue.
    - Other.
4. If the customer selects Other, the system must allow additional comments.
5. The system must validate that a return reason has been selected.
6. The selected reason must be saved with the return request.

Expected Result: Each return request contains a valid and structured return reason.

Priority: High

FR-05: Upload Supporting Images

Description: The system must allow customers to upload product images when supporting evidence is required.

Requirements:

1. The system must provide an image upload option when required by the return policy.
2. The customer must be able to select images from their device.
3. The system must validate supported image formats.
4. The system must validate file size against the configured maximum.
5. The system must enforce the configured maximum number of images.
6. The system must display an error for unsupported files.
7. The customer must be able to preview uploaded images.
8. The customer must be able to remove an image before submission.
9. Uploaded images must be linked to the correct return request.

Expected Result: Customers can provide supporting product images where required.

Priority: Medium

FR-06: Submit Product Return Request

Description: The system must allow customers to submit a return request for an eligible purchased product.

Requirements:

1. The customer must be authenticated.
2. The customer must select an eligible purchased product.
3. The customer must select a return reason.
4. The customer must enter additional comments where required.
5. The system must validate all mandatory information.
6. The system must verify return eligibility before submission.
7. The system must check supporting image requirements.
8. The system must prevent submission if mandatory information is missing.
9. The system must save the submitted return request.
10. The system must generate a unique return request ID.
11. The system must record the submission date and time.

Expected Result: A valid return request is successfully created and stored.

Priority: High

FR-07: Generate Return Request ID

Description: The system must generate a unique reference ID for each successfully submitted return request.

Requirements:

1. The system must generate an ID after successfully saving a request.
2. Each return request ID must be unique.
3. The ID must be associated with the correct customer and order.
4. The ID must be stored in the system.
5. The ID must be displayed in the confirmation message.
6. The ID must be available when the customer views request details.

Expected Result: Every successful return request can be identified using a unique reference ID.

Priority: High

FR-08: Return Request Confirmation

Description: The system must confirm successful return request submission.

Requirements:

1. The system must display a confirmation message after successful submission.
2. The confirmation must include the return request ID.
3. The confirmation must display the relevant product and order reference.
4. The confirmation must display the submission date.
5. The system must send a notification through the configured communication channel, such as email or SMS.
6. The notification must include the return request ID.
7. The system must record the notification delivery status.
8. A notification failure must not delete or invalidate a successfully saved return request.

Expected Result: Customers receive confirmation that their return request has been recorded.

Priority: High

FR-09: View Return Request History

Description: The system must allow customers to view their submitted return requests.

Requirements:

1. The customer must be able to access their return request history after login.
2. The system must display return requests associated with the customer's account.
3. Each request must display its unique request ID.
4. Each request must display the product name.
5. Each request must display the submission date.
6. Each request must display its current status.
7. The system must display a message if no return requests exist.

Expected Result: Customers can view all their submitted return requests.

Priority: Medium

FR-10: View Return Request Details

Description: The system must allow customers to view the complete details of a submitted return request.

Requirements:

1. The customer must be able to select a return request from their history.
2. The system must display the return request ID.
3. The system must display the related order reference.
4. The system must display the product name.
5. The system must display the selected return reason.
6. The system must display additional comments, where available.
7. The system must display the submission date.
8. The system must display uploaded images, where applicable.
9. The system must display the current return request status.
10. The system must prevent customers from viewing another customer's request details.

Expected Result: Customers can review the information associated with their return requests.

Priority: Medium

FR-11: Display Return Request Status

Description: The system must display the current status of each return request.

Requirements:

1. The system must display a status for each submitted return request.
2. The initial status must be Submitted after successful request creation.
3. The system must support agreed statuses such as:
    - Submitted
    - Under Review
    - Approved
    - Rejected
4. Status changes must be recorded when performed by an authorized process or user.
5. The customer must be able to view the latest recorded status.
6. The system must display an appropriate message if status information cannot be retrieved.

Expected Result: Customers can understand the current status of their return requests.

Priority: Medium

6. Non-Functional Requirements

NFR-01: Security

- The system must authenticate customers before allowing access to account information.
- Customers must only access their own orders and return requests.
- Uploaded images and customer information must be protected against unauthorized access.

NFR-02: Performance

- The system should respond to normal customer actions within an acceptable time.
- The system should support multiple customer requests without significant performance degradation.
- Performance targets must be confirmed during technical design.

NFR-03: Usability

- The interface should be simple and easy to understand.
- Validation messages should explain what the customer needs to correct.
- The return submission process should use clear and consistent labels.

NFR-04: Reliability

- Successfully submitted return requests must be stored accurately.
- A notification delivery failure must not result in loss of a saved request.
- The system should handle unexpected errors gracefully.

NFR-05: Maintainability

- Return policy rules and return reasons should be configurable where appropriate.
- The system should support future enhancements.
- Functional components should be documented.

7. Business Rules

Rule ID	Business Rule
BR-01	Only authenticated customers can submit return requests.
BR-02	Only eligible purchased products can be returned.
BR-03	Return eligibility must follow the approved return policy.
BR-04	A return reason is mandatory.
BR-05	Supporting images must be provided when required.
BR-06	Every successful return request must have a unique ID.

BR-07 Customers can only view their own return requests.
BR-08 Return request status must reflect the latest recorded update.

8. Data Requirements

The system should capture the following information:

Field	Description
Customer ID	Unique customer reference
Order ID	Unique order reference
Product ID	Unique product reference
Product Name	Name of purchased product
Return Request ID	Unique return request reference
Return Reason	Selected reason for return
Additional Comments	Customer-provided explanation
Supporting Images	Uploaded image references, where applicable
Submission Date	Date and time of request submission
Return Status	Current recorded request status
Notification Status	Notification delivery result

9. Error Handling and Validation

The system must handle the following scenarios:

Scenario	Expected System Behaviour
Customer is not logged in	Display login requirement
Product is not eligible	Display reason and prevent submission
Return reason is missing	Display validation message
Required image is missing	Prevent submission and display message
Unsupported image format	Display file format error
Image exceeds file size limit	Display file size error
Mandatory information is missing	Highlight missing information
Request cannot be saved	Display an error and avoid showing false confirmation
Notification delivery fails	Preserve request and record notification failure
Customer accesses another customer's request	Deny access

10. Requirement Traceability Matrix

BRD Requirement	Related Functional Requirements
BR-01 View purchased products	FR-02
BR-02 Check return eligibility	FR-03
BR-03 Select return reason	FR-04
BR-04 Submit return request	FR-06
BR-05 Upload supporting images	FR-05
BR-06 Generate unique request ID	FR-07
BR-07 Send confirmation	FR-08
BR-08 View request details and status	FR-09, FR-10, FR-11
BR-09 Analyse return trends	Future reporting phase

11. Assumptions and Dependencies

- Hi Ecommerce is a fictional business.
- Synthetic data will be used for the portfolio project.
- Customer accounts and order details are assumed to be available.
- Return policy rules will be provided by business stakeholders.
- Image upload limits and supported formats will be confirmed during technical design.
- Notification delivery will use a simulated or configured service.
- Actual payment and courier integrations are outside the initial release.

12. Acceptance and Sign-Off

The functional requirements will be considered ready for approval when:

1. All functional requirements have been reviewed.
2. Business rules have been agreed.
3. Requirement priorities have been confirmed.
4. Acceptance criteria are defined for related user stories.
5. Requirements are traceable to the BRD.
6. Business stakeholders have reviewed the document.

Role	Name	Status
Business Sponsor	To be assigned	Pending
Product Owner	To be assigned	Pending
Business Analyst	Portfolio Project Author	Drafted
QA Tester	To be assigned	Pending

Document Status: Draft – For Portfolio Demonstration